

YingShuang Zou

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EDUCATION

Northwestern Polytechnical University - Software Engineering

Sep 2018 - Jul 2022

- GPA: 89.42 (first six terms)
- Comprehensive evaluation: 1/244 (first academic year) 7/244 (second academic year)

SKILLS

- Proficient in using C++, Java and Python for project development, object-oriented programming
- Familiar with Linux, especially the use of Ubuntu;
- Familiar with ROS operating system, able to use OpenCV, Eigen, Gazebo, Docker and other tools;
- Familiar with Git tools, rich experience in team management and cooperative development.
- Proficient in debugger (and IDEs, such as Clion); rich experience in debugging.

RELATED EXPERIENCE

ICRA Robomaster AI Challenge - Project Manager

Dec 2019 - Present

In this competition, we need to use official robots to perceive information from battlefield, make autonomous decisions, navigate, and launch projectiles to hit enemy robots for shooting confrontation.

- As a project manager, I am responsible for managing the team, scheduling the competition, dividing a project into technical modules, and arranging tasks in time;
- During the COVID-19, I wrote a Gazebo simulation platform for the competition which can simulate various sensor information, so that we can test the upper strategy conveniently without a real robot.
- Responsible for mapping, localization, and planning part for the autonomous wheel robot, and in this project we use the Cartographer method to map and get accurate position of robots.
- Modifying the global-planning algorithm by inputting the path of another robot into the costmap of the current robot, so that simple multi robot planning is realized.

Soccer Training Assistant System Based on UAV (Excellent Conclusion of National College Students' Innovation Training Projects) - Major Member

Apr 2020 - Apr 2021

The purpose of this project is to establish a UAV system that can assist soccer training, which can record the game autonomously and display the information of court.

- As the main person in charge, I established a whole system framework, divided each module, and arranged the time and tasks.
- Participate in the construction of UAV, related implementation of obstacle avoidance module, communication module, and debugging. These functions are the basis of UAV Systems.
- Complete the object detection algorithm module and related interface implementations. This project uses the pre-trained YOLOV5-s model and then trains it in soccer field datasets to achieve high precision.

FIRA Roboworld Cup Robosot Race Development Group - Team Leader

Jun 2019 - Aug 2019

This is a small-sized soccer robot competition for wheeled platforms. After the game starts, the robot must search for the color ball in the field and push the ball into the corresponding color area to get the score.

- Responsible for upper strategy modules of wheeled robots. The implementation of upper strategy uses the form similar to FSM to complete the transition between different states according to the current information.
- Participate in the robot sensor drive module (like publishing IMU data in the form of ROS topic).
- Responsible for the implementation of navigation and obstacle avoidance functions for the wheeled robot.

HONORS & AWARDS

2019&2020 Outstanding students of Northwestern Polytechnic University (First Class Scholarship)

2019&2020 Wu Yajun scholarship

3rd prize in 2021 University AI Challenge (hosted by IEEE ICRA and Robomaster) June 2021

2nd place in FIRA Roboworld Cup, Robosot Race Competition Development Group Aug 2019

1st place in China Robotics and Artificial Intelligence Competition, Microsoft Simulation Football Tournament Oct 2019

6th (in 72 terms) in ICRA Robomaster AI CHALLENGE Aug 2020

And other competition awards.

SCHOOL ACTIVITIES

Student Tennis Association of Northwestern Polytechnical University -
President

Aug 2020 - Present

- Successfully held the 8th "SanHang Cup" tennis team competition, more than 14 colleges participated in the competition, and attracted many teachers and students who love tennis participate actively;
- Responsible for the daily management and training arrangements of the Tennis Association, helping the development of sports, assisting the Sports Department and Sports Association, providing the opportunity to experience tennis for the whole university.

Homework Guidance for School Student - Volunteer

Feb 2020 - Jun 2020

- During the COVID-19, provide homework guidance and psychological comfort for the children whose parents are fighting the epidemic.

Technical Support for Online Teaching - Volunteer

Mar 2020 - May 2020

- Mainly responsible for assisting the professor of "China's Modern History Outline", assisting using online platforms, resolving and feedback related technical issues, and volunteering for over 50 hours.